



MOLECULAR BIOLOGY AND GENETICS: LABORATORY REPORT

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Abstract

This study investigates the biological impact of Ultraviolet (UV) radiation on bacterial genomic integrity through a comprehensive molecular biology workflow. The experimental procedure began with the cultivation of bacteria in liquid MRS medium, followed by controlled UV exposure for 30 minutes to induce DNA damage. To assess cellular recovery and genetic stability, DNA was extracted using the Trizol-chloroform method across varying post-exposure incubation intervals. The extracted genetic material was subsequently analyzed via Polymerase Chain Reaction (PCR) and Agarose Gel Electrophoresis to evaluate the success of DNA recovery and the extent of UV-induced modifications. This integrated approach provides a practical framework for understanding DNA damage mechanisms and the fundamental techniques of genetic analysis, bridging the gap between theoretical molecular biology and laboratory application.

Keywords: UV radiation; DNA extraction; Trizol method; PCR; Electrophoresis; Bacterial DNA integrity.

1. Introduction

In the field of molecular biology, the ability to isolate, amplify, and analyze genetic material is a foundational requirement for both research and diagnostic applications. Among the various environmental factors affecting genetic stability, Ultraviolet (UV) radiation

serves as a primary mutagenic agent capable of inducing structural lesions in DNA. Understanding how these stressors impact microbial genomes is essential for studying DNA repair mechanisms and evolutionary biology.

This laboratory series was designed to equip students with essential techniques in molecular genetics while observing the real-time effects of UV-induced stress on bacterial cultures. The workflow encompasses a complete cycle of molecular analysis:

- **Bacterial Cultivation:** Establishing a robust biomass in MRS medium at 37°C.
- **Mutagenic Treatment:** Subjecting the samples to UV exposure and monitoring recovery phases.
- **Genomic Extraction:** Utilizing the Trizol reagent protocol to isolate high-quality DNA through organic phase separation and centrifugation at 12000g.
- **Molecular Characterization:** Employing PCR amplification and agarose gel electrophoresis at 110mV to visualize and verify the targeted DNA segments.

By integrating these core methodologies, the experiment provides a comprehensive assessment of the impact of physical stressors on bacterial DNA, reinforcing the theoretical principles of molecular genetics through empirical evidence.

2. Theoretical Background

2.1. UV-Induced DNA Damage and Repair

Ultraviolet (UV) radiation, particularly in the UV-C (200–280 nm) and UV-B (280–320 nm) ranges, is a potent mutagen that causes direct damage to DNA. The most common lesions are cyclobutane pyrimidine dimers (CPDs) and 6-4 photoproducts, formed between adjacent pyrimidine bases (cytosine or thymine) [1]. These structural distortions interfere with DNA replication and transcription, potentially leading to mutations or cell death. Bacteria have evolved several repair mechanisms, including photoreactivation and nucleotide excision repair (NER), to maintain genomic integrity following UV exposure [1].

2.2. Principles of Nucleic Acid Extraction

The isolation of high-purity genomic DNA is a prerequisite for downstream molecular applications. The Trizol method (acid guanidinium thiocyanate-phenol-chloroform extraction) is a widely used technique for the simultaneous isolation of RNA, DNA, and proteins [2]. Guanidinium thiocyanate acts as a powerful denaturant that lyses cells and inactivates nucleases. The addition of chloroform leads to phase separation: DNA partitions into the interphase or organic phase under acidic conditions, but can be recovered from the organic phase using specialized back-extraction buffers (BEB) which shift the pH to allow DNA solubilization into the aqueous phase [2].

2.3. Polymerase Chain Reaction (PCR)

PCR is an *in vitro* technique for the exponential amplification of specific DNA sequences. The process relies on thermal cycling, consisting of three discrete temperature steps: denaturation (to separate DNA strands), annealing (to allow primers to bind to target sequences), and extension (where a thermostable DNA polymerase, such as Taq, synthesizes new DNA strands) [3]. PCR allows for the detection of even minute quantities of genetic material, making it essential for verifying the presence and integrity of extracted DNA.

2.4. Agarose Gel Electrophoresis

Agarose gel electrophoresis is the standard method for separating DNA fragments based on their size and charge. Under an electric field, negatively charged DNA molecules migrate through the porous agarose matrix toward the positive anode [4]. Smaller fragments migrate faster than larger ones, allowing for the determination of fragment sizes when compared against a molecular weight marker (ladder). Visualization is typically achieved using fluorescent dyes, such as GelRed, which intercalate with DNA and fluoresce under UV light [4].

3. Materials & Methods

4. Results & Discussion

5. Conclusion

References

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